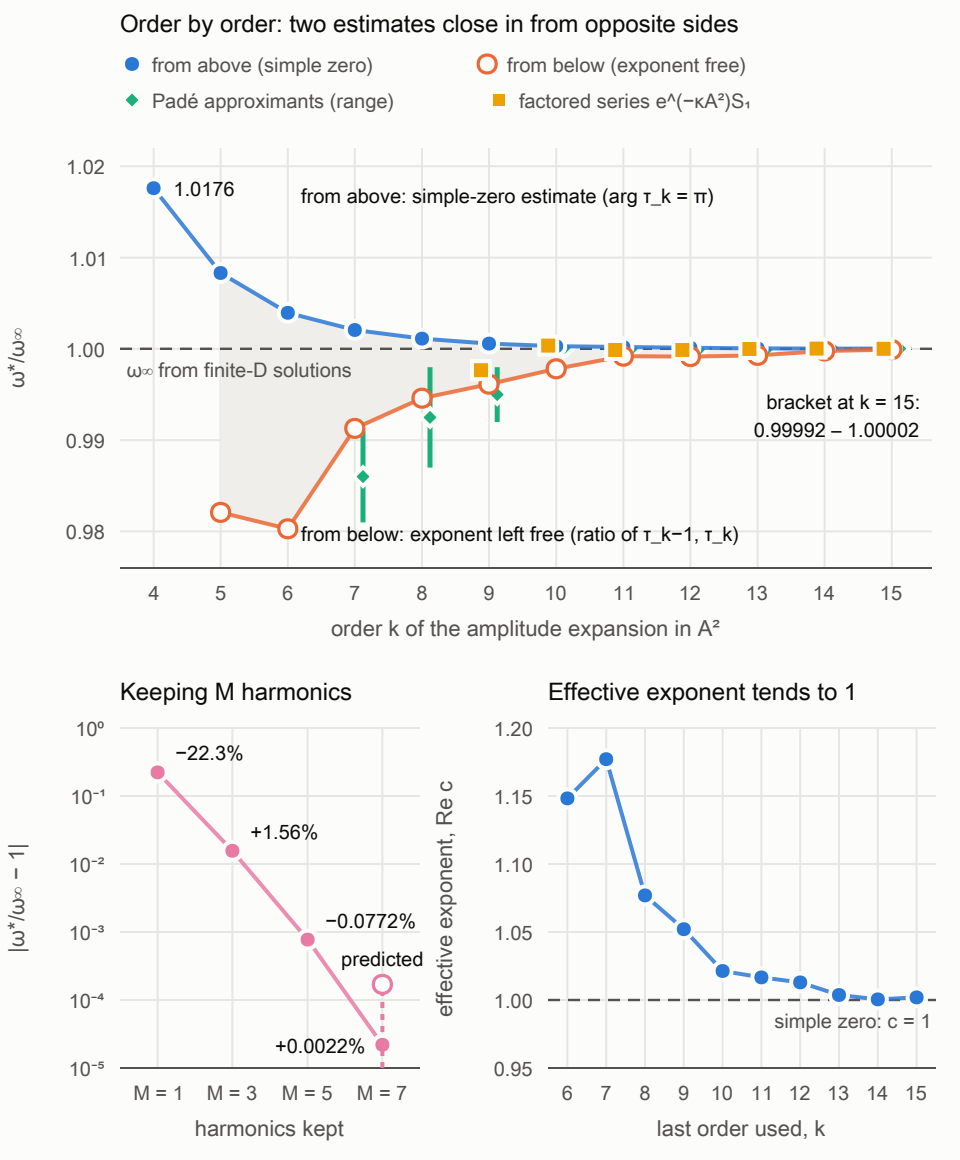
Order by order: two estimates close in from opposite sides
from above (simple zero)
from below (exponent free)
Padé approximants (range)
factored series e^(−κA²)S₁
1.02
1.01
1.00
0.99
0.98
ω*/ω∞
1.0176
from above: simple-zero estimate (arg τ_k = π)
ω∞ from finite-D solutions
bracket at k = 15:
0.99992 – 1.00002
from below: exponent left free (ratio of τ_k−1, τ_k)
4
5
6
7
8
9
10
11
12
13
14
15
order k of the amplitude expansion in A²
Keeping M harmonics
10⁰
10⁻¹
10⁻²
10⁻³
10⁻⁴
10⁻⁵
|ω*/ω∞ − 1|
−22.3%
+1.56%
−0.0772%
predicted
+0.0022%
M = 1
M = 3
M = 5
M = 7
harmonics kept
Effective exponent tends to 1
1.20
1.15
1.10
1.05
1.00
0.95
effective exponent, Re c
simple zero: c = 1
6
7
8
9
10
11
12
13
14
15
last order used, k